

# Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats

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Technology Executives & Network Infrastructure Leaders

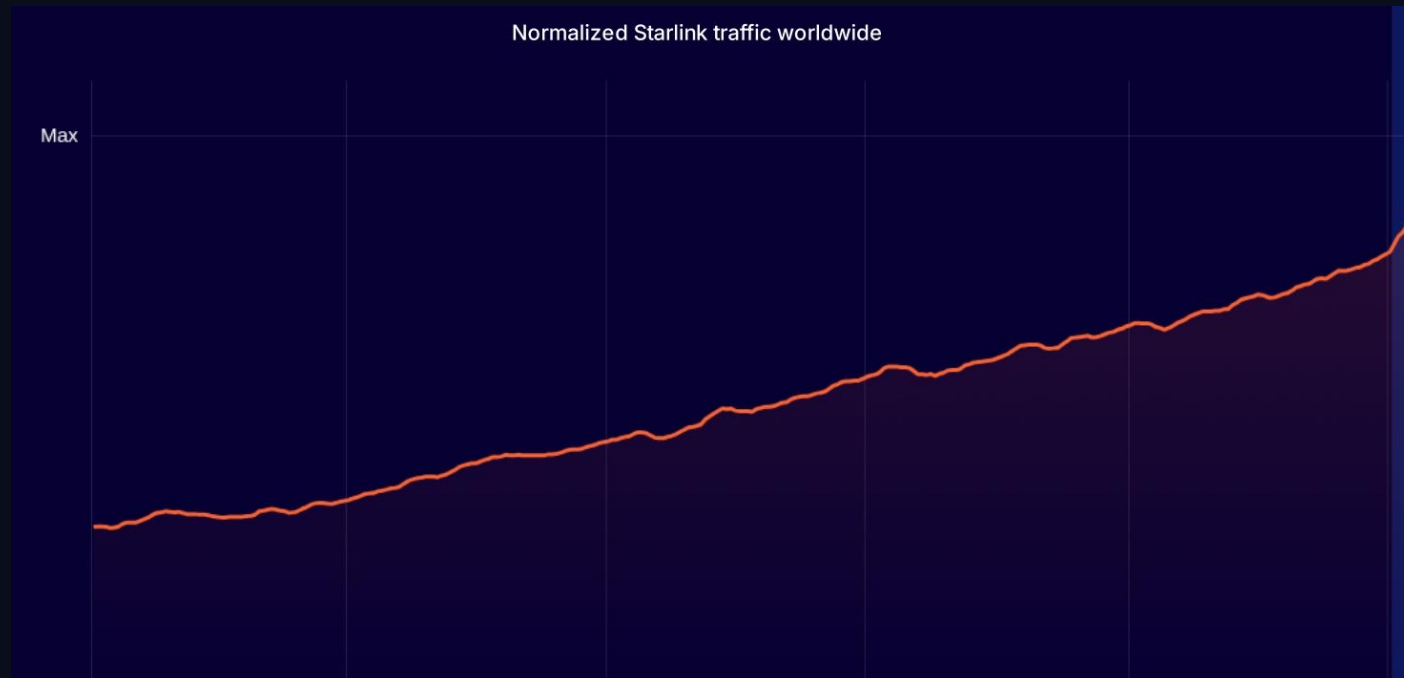
# 17% Global Traffic Growth and Starlink's 3.3x Surge

## +17%

Year-over-year growth in global Internet traffic, continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.

## 3.3x

SpaceX Starlink traffic surge during 2024, bringing satellite Internet connectivity to previously underserved locations. Trend normalized relative to peak observed request volume.



# Google Leads Top 10 Services as Generative AI Rises

## Top 10 Most Popular Internet Services

- #1 Google
- #2 Facebook
- #3 Apple
- #4 TikTok
- #5 AWS
- #6 Microsoft
- #7 Instagram
- #8 YouTube
- #9 Amazon
- #10 WhatsApp

| Rank | Service   |
|------|-----------|
| #1   | Google    |
| #2   | Facebook  |
| #3   | Apple     |
| #4   | TikTok    |
| #5   | AWS       |
| #6   | Microsoft |
| #7   | Instagram |
| #8   | YouTube   |
| #9   | Amazon    |
| #10  | WhatsApp  |

## Generative AI on the Rise

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background. Rankings based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

# Post-Quantum Encryption Jumps from 2% to 13% of TLS



**2% → 13%**

Post-quantum encryption share of TLS 1.3 traffic after Chrome 124 enabled it by default on April 17, 2024. Firefox and Safari have also begun rollout.

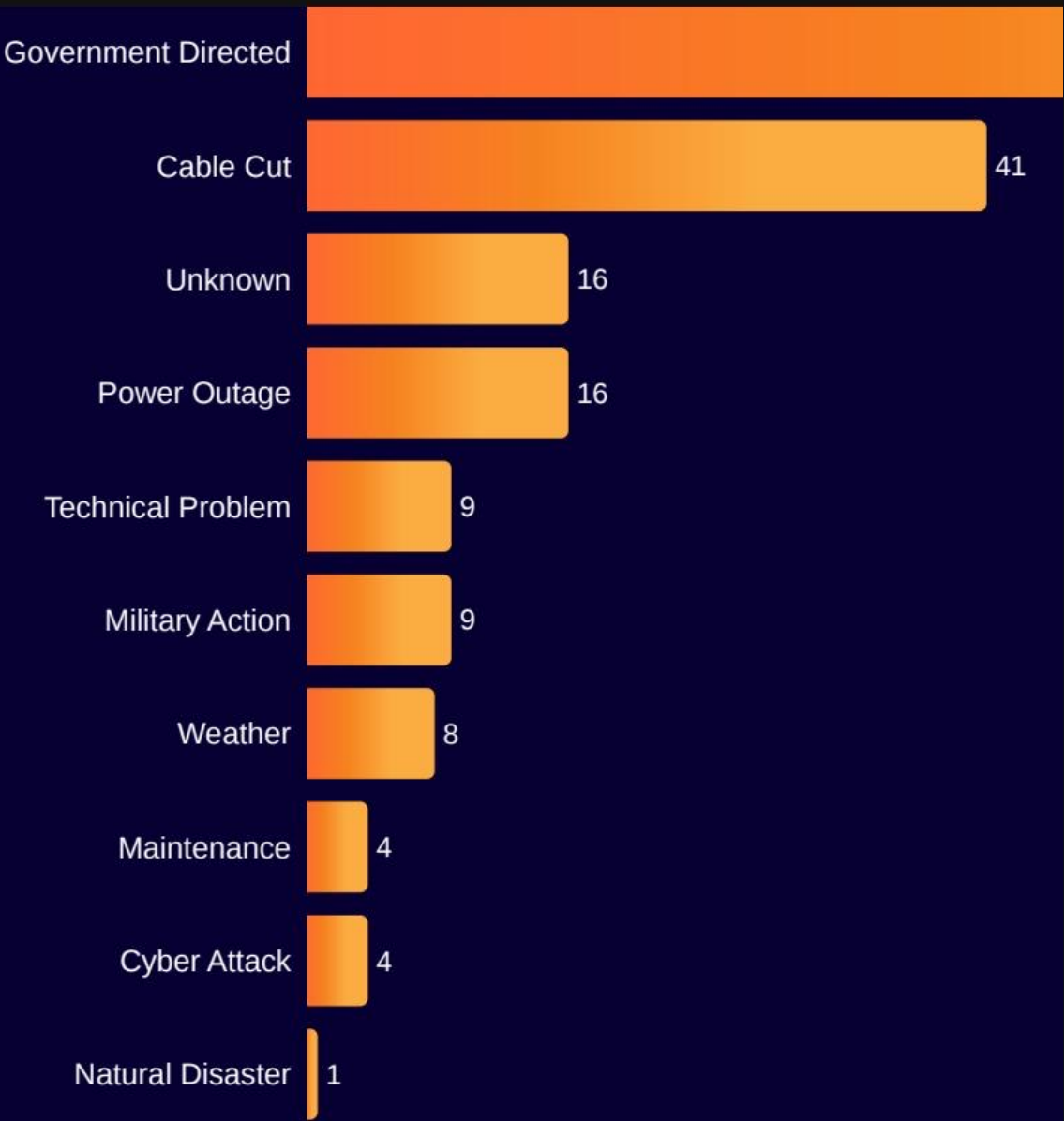
## HTTP/3: 21% Adoption

The latest protocol version running on QUIC, providing faster connections, packet-loss mitigation, and encryption by default.

## Go Becomes Top API Client Language

Go leads with 12% of automated API requests, followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).

# 225 Major Disruptions: Government Shutdowns Lead



| Causes of Major Internet Disruptions (2024) |      |
|---------------------------------------------|------|
| Government Directed                         | 116+ |
| Cable Cut                                   | 41   |
| Unknown                                     | 16   |
| Power Outage                                | 16   |
| Technical Problem                           | 9    |
| Military Action                             | 9    |
| Weather                                     | 8    |
| Maintenance                                 | 4    |
| Cyber Attack                                | 4    |
| Natural Disaster                            | 1    |
| Fire                                        | 1    |

# 6.5% of Traffic Mitigated: Persistent Threats and Bot

## 6.5%

of all traffic processed by Cloudflare was mitigated, indicating the massive scale of ongoing threats.

## Gambling & Games

The most attacked industry category observed across Cloudflare's global network in 2024.

## 68%

of global bot traffic originated from just the top 10 countries, showing extreme geographic concentration.

## Log4j Persistence

The Log4j vulnerability (CVE-2021-44228), first disclosed in December 2021, continues to be actively targeted by attackers more than two years later — underscoring the persistent nature of widely known vulnerabilities.

## Email Security Threats

4.3% of emails processed were malicious. Of those, 43% contained a deceptive link designed to trick recipients. The .bar TLD originated the largest share of malicious and spam email observed during the year.

# Five Takeaways Shaping the 2025 Internet Outlook

## 01 Traffic Growth Trajectory

Global Internet traffic grew 17% year-over-year, with Starlink traffic surging 3.3x as satellite connectivity expanded to underserved regions.

## 02 Post-Quantum Momentum

Post-quantum encryption adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

## 03 Connectivity Fragility

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed, followed by cable cuts (41) and power outages (16).

## 04 Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

## 05 Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client language, capturing roughly 12% of automated API traffic.